

Arch Linux Btrfs Installation Guide

Manual filesystem setup - UEFI + Btrfs subvolumes + /boot + fstab

Layout used in this guide

```
@           -> /
@home       -> /home
@log        -> /var/log
@pkg        -> /var/cache/pacman/pkg
@swap       -> /swap
@snapshots  -> /.snapshots
```

```
EFI         -> /boot
```

Important: The example commands use /dev/nvme0n1p1 for EFI and /dev/nvme0n1p2 for Btrfs root. Replace them with your actual partitions. Formatting the wrong partition destroys its existing data.

1. Identify the partitions

Start by displaying the disks, partitions, filesystem types, labels, and mount points. Identify the EFI System Partition and the partition you intend to use for Arch's Btrfs filesystem.

```
lsblk -f
```

For the examples in this guide:

```
/dev/nvme0n1p1 = EFI
```

```
/dev/nvme0n1p2 = Btrfs root
```

Do not continue until you are certain which partitions are correct.

2. Format the EFI partition

Format the EFI System Partition as FAT32. Skip this formatting command if you intentionally need to preserve an existing shared EFI partition.

```
mkfs.fat -F 32 /dev/nvme0n1p1
```

3. Format the root partition as Btrfs

Create the Btrfs filesystem on the root partition.

```
mkfs.btrfs -f /dev/nvme0n1p2
```

Verify the result before continuing.

4. Verify the filesystems

Check the filesystem types again. The EFI partition should show vfat and the root partition should show btrfs.

```
lsblk -f
```

5. Mount the Btrfs top level

Mount the newly created Btrfs filesystem temporarily at /mnt. This exposes the Btrfs top level so the subvolumes can be created.

```
mount /dev/nvme0n1p2 /mnt

findmnt /mnt
```

6. Create the Btrfs subvolumes

Create the six subvolumes used by this layout.

```
btrfs subvolume create /mnt/@
btrfs subvolume create /mnt/@home
btrfs subvolume create /mnt/@log
btrfs subvolume create /mnt/@pkg
btrfs subvolume create /mnt/@swap
btrfs subvolume create /mnt/@snapshots
```

Verify that all six were created:

7. Verify the subvolumes

The output should contain @, @home, @log, @pkg, @swap, and @snapshots.

```
btrfs subvolume list /mnt
```

8. Unmount the Btrfs top level

The temporary top-level mount is no longer needed. Unmount it before mounting @ as the installed system's root.

```
umount /mnt
```

9. Mount the root subvolume

Mount @ at /mnt. Compression is enabled with zstd and noatime avoids routine access-time writes.

```
mount -o subvol=@,compress=zstd,noatime \
/dev/nvme0n1p2 /mnt

findmnt /mnt
```

The mount options should indicate the @ subvolume (commonly shown as subvol=/@).

10. Create the mount-point directories

Create the directories where the remaining Btrfs subvolumes and EFI partition will be mounted.

```
mkdir -p /mnt/home
mkdir -p /mnt/var/log
mkdir -p /mnt/var/cache/pacman/pkg
mkdir -p /mnt/swap
mkdir -p /mnt/.snapshots
mkdir -p /mnt/boot
```

11. Mount @home

Mount the home subvolume.

```
mount -o subvol=@home,compress=zstd,noatime \
/dev/nvme0n1p2 /mnt/home
```

12. Mount @log

Keep /var/log outside root snapshots.

```
mount -o subvol=@log,compress=zstd,noatime \
/dev/nvme0n1p2 /mnt/var/log
```

13. Mount @pkg

Keep Pacman's package cache outside root snapshots.

```
mount -o subvol=@pkg,compress=zstd,noatime \
/dev/nvme0n1p2 /mnt/var/cache/pacman/pkg
```

14. Mount @snapshots

Mount the dedicated Snapper snapshot subvolume at /.snapshots.

```
mount -o subvol=@snapshots,compress=zstd,noatime \
/dev/nvme0n1p2 /mnt/.snapshots
```

15. Mount @swap

Mount the dedicated swap subvolume. Compression is intentionally omitted because this location will later contain the Btrfs swapfile.

```
mount -o subvol=@swap,noatime \
/dev/nvme0n1p2 /mnt/swap
```

16. Mount the EFI System Partition

Mount the EFI partition at /boot.

```
mount /dev/nvme0n1p1 /mnt/boot
```

17. Verify the complete mount layout

Before installing Arch, inspect every mount. This is the main sanity check.

```
findmnt -R /mnt -o TARGET,SOURCE,FSTYPE,OPTIONS
```

Confirm the mapping:

```
/mnt -> @
/mnt/home -> @home
/mnt/var/log -> @log
/mnt/var/cache/pacman/pkg -> @pkg
/mnt/swap -> @swap
/mnt/.snapshots -> @snapshots
/mnt/boot -> EFI partition
```

18. Install the base system

Once the mount layout is correct, install the Arch base system and Btrfs utilities.

```
pacstrap -K /mnt \
  base \
  linux \
  linux-firmware \
  btrfs-progs
```

Add any other packages you want to install at this stage as needed.

19. Generate /etc/fstab

After pacstrap creates /mnt/etc, generate fstab using UUIDs.

```
genfstab -U /mnt > /mnt/etc/fstab
```

20. Inspect fstab

Do not continue without reading the generated file. Confirm entries exist for root, home, log, package cache, swap directory, snapshots, and /boot.

```
cat /mnt/etc/fstab
```

The Btrfs entries should contain the appropriate subvol= options. @swap being mounted at /swap is not the same thing as having active swap; the actual swapfile is created later.

21. Enter the installed system

Once the filesystem and fstab look correct, enter the new Arch installation.

```
arch-chroot /mnt
```

Final filesystem layout

```
Btrfs filesystem
|
|-- @          -> /
|-- @home      -> /home
|-- @log       -> /var/log
|-- @pkg       -> /var/cache/pacman/pkg
|-- @swap      -> /swap
`-- @snapshots -> /.snapshots

EFI System Partition
`-- /boot
```

Next steps after arch-chroot typically include timezone, locale, hostname, networking, users/sudo, swapfile and hibernation configuration, initramfs configuration, and bootloader installation.